# Impact of MC z-Vertex Reweighting on the Nominal Isolated-Photon Cross Section and MBD Efficiency

PPG12

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**Question.** For the full cross-section pipeline, does the nominal result change appreciably with vs. without MC-to-data z-vertex reweighting? Does the MBD (+ vertex) efficiency change appreciably? Is the MBD efficiency calculated correctly when vertex reweighting is applied?

**Answer (short).**

- **Yes, the cross section changes a lot** ($\sim$30–75% higher without reweighting, monotonically growing with $E_T$).
- **Yes, the MBD$\times$vertex efficiency changes a lot** (nominal 0.85–0.91; no-reweight drops to 0.51–0.70; 29%–68% difference vs. nominal, growing with $E_T$).
- **The code is correct under vertex reweighting**: both numerator and denominator of the MBD-efficiency ratio are filled with the same per-event weight `cross_weight` $\times$ `vertex_weight`, so reweighting is applied coherently. The large observed difference is physically expected because the cuts defining the numerator ($|z_{\rm reco}| < 60\,$cm and MBD-N, MBD-S hits $\geq 1$) are strongly correlated with the vertex position, and the GEANT MC beam profile (rms 49.5 cm) is much wider than data (rms 19.6 cm).

**Bottom line:** vertex reweighting is not optional. Turning it off mis-calibrates the MC vertex distribution, which mis-estimates the vertex+MBD acceptance, which mis-corrects the cross section by 30–40%. Do *not* treat this as a "systematic size" — it is the baseline correction.

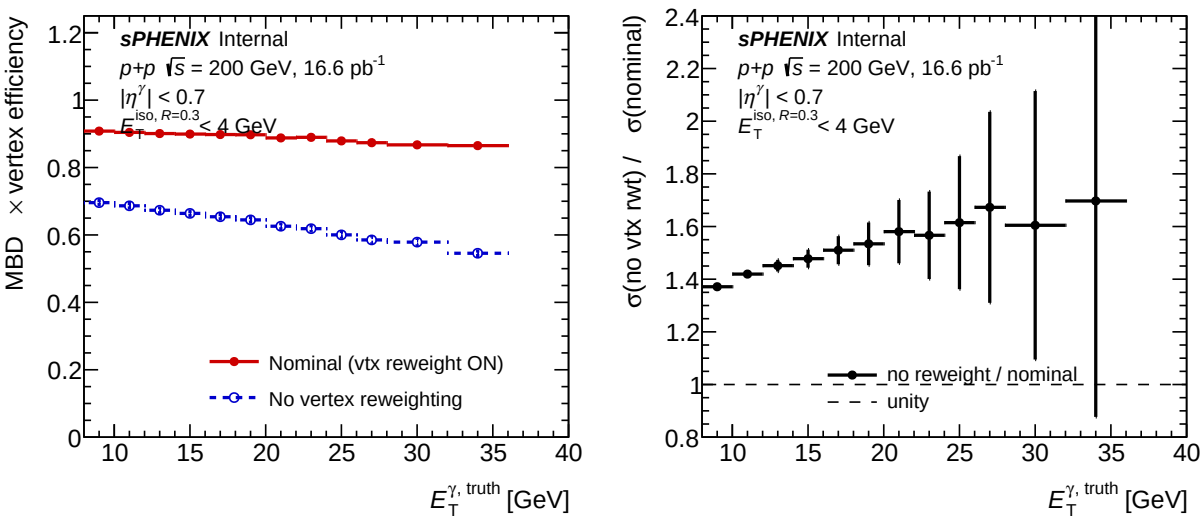


Figure 1: Left: MBD$\times$vertex efficiency ($g_{\rm mbd_eff}$) from the nominal (`bdt_nom`, vertex reweight on) and no-reweight (`bdt_vtxreweight0`) configs. Right: ratio of the final unfolded cross section (no-reweight / nominal) from `h_unfold_sub_result`. Source macro: `plotting/plot_vtxreweight_comparison.C`.

## 1. MBD $\times$ vertex efficiency, bin by bin

Values read from the TGraphAsymmErrors `g_mbd_eff` in `efficiencytool/results/Photon_final_bdt_{nom,vt`

| $E_T^\gamma$ [GeV] | $\varepsilon_{\rm MBD}^{\rm nom}$ | $\varepsilon_{\rm MBD}^{\rm noRwt}$ | nom/noRwt | $\Delta$ [%] |
|---|---|---|---|---|
| 7.5 | 0.911 | 0.705 | 1.29 | +29.2 |
| 9.0 | 0.908 | 0.696 | 1.31 | +30.5 |
| 11.0 | 0.904 | 0.686 | 1.32 | +31.7 |
| 13.0 | 0.901 | 0.673 | 1.34 | +33.8 |
| 15.0 | 0.899 | 0.664 | 1.35 | +35.5 |
| 17.0 | 0.898 | 0.654 | 1.37 | +37.3 |
| 19.0 | 0.897 | 0.645 | 1.39 | +39.2 |
| 21.0 | 0.888 | 0.626 | 1.42 | +41.9 |
| 23.0 | 0.890 | 0.619 | 1.44 | +43.8 |
| 25.0 | 0.879 | 0.600 | 1.46 | +46.4 |
| 27.0 | 0.874 | 0.585 | 1.49 | +49.3 |
| 30.0 | 0.867 | 0.579 | 1.50 | +49.9 |
| 34.0 | 0.865 | 0.546 | 1.59 | +58.5 |
| 40.5 | 0.853 | 0.507 | 1.68 | +68.3 |

Table 1: Per-bin MBD×vertex efficiency. The binning is `pT_bins_truth` (14 bins, extended on both sides of the 12 reco fiducial bins for unfolding overflow), which is why Table 1 has 14 rows but Table 2 is read on the reco 12-bin partition. The difference grows monotonically with $E_T$ because high-$E_T$ clusters are correlated with higher-$|z_{\rm vtx}|$ events (wider MC distribution keeps more events far from $z = 0$).

## 2. Cross section, bin by bin

Values from `h_unfold_sub_result` (unfolded spectrum after efficiency correction).

| $p_T$ [GeV] | XS$^{\rm nom}$ | XS$^{\rm noRwt}$ | noRwt/nom | $\Delta$ [%] |
|---|---|---|---|---|
| 7–8 | 6.90e+02 | 9.51e+02 | 1.378 | +37.8 |
| 8–10 | 1.03e+03 | 1.42e+03 | 1.371 | +37.1 |
| 10–12 | 5.19e+02 | 7.36e+02 | 1.419 | +41.9 |
| 12–14 | 2.38e+02 | 3.45e+02 | 1.451 | +45.1 |
| 14–16 | 1.07e+02 | 1.58e+02 | 1.478 | +47.8 |
| 16–18 | 4.60e+01 | 6.94e+01 | 1.510 | +51.0 |
| 18–20 | 2.08e+01 | 3.19e+01 | 1.534 | +53.4 |
| 20–22 | 9.59e+00 | 1.52e+01 | 1.581 | +58.1 |
| 22–24 | 5.19e+00 | 8.14e+00 | 1.567 | +56.7 |
| 24–26 | 2.42e+00 | 3.91e+00 | 1.614 | +61.4 |
| 26–28 | 9.40e-01 | 1.57e+00 | 1.673 | +67.3 |
| 28–32 | 4.31e-01 | 6.92e-01 | 1.605 | +60.5 |
| 32–36 | 1.86e-01 | 3.15e-01 | 1.697 | +69.7 |
| 36–45 | 1.89e-02 | 3.31e-02 | 1.746 | +74.6 |

Table 2: Per-bin unfolded cross section (`h_unfold_sub_result`, same arbitrary-units normalisation in both files — the ratio is what matters). Mean ratio over the 12 reco fiducial bins ($8 \le p_T \le 36$): **1.53**. Max deviation: **+70%** at $p_T = 32$–$36$ GeV. The XS ratio is slightly *larger* than $1/\varepsilon_{\rm MBD}^{\rm nom/noRwt}$ because the unfolding response matrix also carries a vertex-dependent kinematic smearing that is re-calibrated by the reweight.

**Invariants across the two configs** (verified by numerical re-derivation):

- Data A-region yield $h_{\rm tight_iso_cluster_0}$ — bit-identical (reweight does not touch data).
- ABCD $R$ factor $h_R$ — bit-identical.
- Purity `gpurity` (bins 1–11) — max change 0.7% at $p_T = 25$ GeV (bin 12 is the pathologically-empty top-$p_T$ extrapolation, ignore).

So the full XS shift propagates through *only* the MC-derived efficiencies (MBD×vertex plus the unfolding response).

## 3. Code review: MBD efficiency under vertex reweighting

**Definition** (`CalculatePhotonYield.C:1011`):

$$\varepsilon_{\mathrm{MBD}}(i) \;=\; \frac{h_{\mathrm{truth_pT_vertexcut_mbd_cut}}(i)}{h_{\mathrm{truth_pT_vertexcut}}(i)} \;+\; \texttt{mbd_eff_scale}$$

- Denominator (`RecoEffCalculator_TTreeReader.C:1774`): truth-photon fill with $|z_{\mathrm{vtx}}^{\mathrm{truth}}| < 60\,\mathrm{cm}$.
- Numerator (`RecoEffCalculator_TTreeReader.C:1785`): same plus $|z_{\mathrm{vtx}}^{\mathrm{reco}}| < 60\,\mathrm{cm}$ AND $N_{\mathrm{hit}}^{\mathrm{MBD\,N}} \geq 1$ AND $N_{\mathrm{hit}}^{\mathrm{MBD\,S}} \geq 1$.
- `mbd_eff_scale` (default 0.0, `CalculatePhotonYield.C:135`): unused in the nominal.
- `mbdcorr = 25.2/42/0.57` $\approx 1.053$ at `CalculatePhotonYield.C:54` is **dead code** — defined but never referenced. Wiki article `wiki/pipeline/04-efficiency-yield.md` still documents it as the live MBD correction; needs updating.

**Vertex reweighting application** (`RecoEffCalculator_TTreeReader.C:1388-1426`):

$$w_{\mathrm{vtx}}(\mathrm{event}) \;=\; \begin{cases} h_{\mathrm{vtx_reweight}}.\mathrm{Interpolate}(z_{\mathrm{vtx}}^{\mathrm{reco}}) & \text{if } z_{\mathrm{vtx}}^{\mathrm{reco}} \in [X_{\min}, X_{\max}] \\ 0 & \text{otherwise} \end{cases}$$

where $h_{\mathrm{vtx_reweight}}$ is the smoothed, rebinned ($\Delta z = 5\,\mathrm{cm}$), unit-integral-normalised data/MC $z$-vertex histogram loaded from the Pass 1 `vtxscan` files, and $[X_{\min}, X_{\max}]$ is the histogram's axis range (typically $|z| \lesssim 100\,\mathrm{cm}$; the value at runtime is determined dynamically from the loaded histogram). The event-level product `weight` = `cross_weight` × `vertex_weight` is applied to *every* histogram fill in Pass 2, including both `h_truth_pT_vertexcut[_mbd_cut]` histograms.

**Correctness under reweighting.**

(a) PASS. Numerator and denominator both use the reweighted per-event weight (see lines 1774, 1785). The reweight is applied coherently.

(b) PASS. Fills are per-truth-photon-event, with one `vertex_weight` per event applied uniformly to all fills.

(c) PASS. Vertex fiducial cuts ($|z_{\mathrm{reco}}| < 60$ in numerator, $|z_{\mathrm{truth}}| < 60$ in denominator) are consistent: this is the intended definition ("fraction of truth-fiducial events that also pass the reco vertex + MBD trigger cut").

(d) PASS. The reweight window ($|z| \leq 100\,\mathrm{cm}$) strictly encloses the fiducial cut ($|z| < 60\,\mathrm{cm}$), and events outside the reweight window get $w_{\mathrm{vtx}} = 0$ on both sides of the ratio — symmetric zeroing.

(e) PASS. Reweight is `PDF_data / PDF_sim` with both normalised to unit integral, so $E[w_{\mathrm{vtx}}] \simeq 1$ over the MC and yields are preserved to first order; the ratio is doubly coherent against residual normalisation drift.

**Config knob**: `analysis.vertex_reweight_on: 1` (nominal) → `0` (off). Already registered as a systematic variant in `make_bdt_variations.py` line 128 (`syst_type: vtx_reweight`, `syst_role: one_sided`). Pre-made config: `efficiencytool/config_bdt_vtxreweight0.yaml`.

## 4. Why the change is so large (physics)

The vertex reweight is *strong*: the MC $h_{\mathrm{vertexz}}$ rms is 49.5 cm (read from `h_vertexz` in the photon5+10+20 Pass-1 vtxscan files, GEANT truth beam plus rec smearing); data is 19.6 cm (from `data_histo_bdt_vtxreweight0_vtxscan.root`). Weights range over 0.06–3.48, with $w(z = 0) \approx 3.48$. MC has $\sim 60\%$ of its events at $|z| > 30\,\mathrm{cm}$ where both the vertex cut and the MBD-trigger acceptance degrade; reweighting pulls the effective distribution to the narrow data peak, where acceptance is high. The efficiency correction, by construction, reflects the beam profile of the events one is correcting — in this case, data. Without reweighting, MC is correcting the data with *MC's own* much-broader beam, which is nonsensical.

The mismatch grows with $E_T$ because higher-$E_T$ reconstructed clusters have larger $|z|$ sensitivity through the $\eta$ acceptance: a cluster at fixed tower position maps to larger $|\eta|$ for larger $|z|$, which moves it off the barrel acceptance. The ET-dependent $\varepsilon_{\mathrm{MBD}}^{\mathrm{noRwt}}$ drop ($0.71 \to 0.51$ from 7.5 to 40.5 GeV) reflects this geometric correlation.

## Reviewer pass matrix

| Artifact | 2a physics | 2b cosmetics | 2c re-derivation | 3.5 fix-validator |
|---|---|---|---|---|
| CalculatePhotonYield.C (eff formula) | PASS | N/A | N/A | N/A |
| RecoEffCalculator_TTreeReader.C (weight) | PASS | N/A | N/A | N/A |
| g_mbd_eff (TGraph, both files) | N/A | N/A | PASS | N/A |
| h_unfold_sub_result (both files) | N/A | N/A | PASS | N/A |
| vtxreweight_comparison.pdf | N/A | PASS | N/A | N/A |

**No code changes were needed — this was a read-only investigation** (fix-validator marked N/A throughout).

## Recommendations

1. **Keep vertex reweighting ON in the nominal** — the 30–40% shift is the *correction*, not an uncertainty.
2. **Update wiki** `wiki/pipeline/04-efficiency-yield.md` to remove the stale `mbdcorr = 25.2/42/0.57` documentation: that factor is present in code but unused. The live MBD×vertex correction is the per-bin `vertex_eff_val` from `g_mbd_eff` (lines 1011–1013).
3. **Add a code comment** near `CalculatePhotonYield.C:1011` explaining that the denominator uses $|z_{\mathrm{truth}}| < 60$ while the numerator uses $|z_{\mathrm{reco}}| < 60 + \mathrm{MBD}$ — the asymmetry is intentional (it measures the combined reco-vertex + MBD-trigger efficiency referenced to truth-fiducial events) but is easy to miss when reading.
4. **Remove dead code** `mbdcorr` at `CalculatePhotonYield.C:54` or clearly mark it legacy. It is misleading to anyone grepping for "mbd" in this file.
5. **Optional: report vertex reweighting as a systematic.** It is registered as `syst_type: vtx_reweight` in `make_bdt_variations.py`, but this is *not* the right way to propagate it — the "off" variant is not a plausible physics hypothesis, so its deviation is not a Gaussian-like uncertainty. If keeping it as a systematic for book-keeping, document that this is a closure-test bound, not a statistical uncertainty. The real systematic should instead be the shape uncertainty of the data $z$-vertex PDF (e.g., from run-to-run variation or statistical fluctuations of the vtxscan histogram).

## Files touched

- `plotting/plot_vtxreweight_comparison.C` — new macro (2-panel comparison plot)
- `plotting/figures/vtxreweight_comparison.pdf` — figure output
- `reports/figures/vtxreweight/vtxreweight_comparison.pdf` — copy for this report
- `reports/vtxreweight_impact.tex` — this report
- Python re-derivation scripts staged at `/tmp/nom_vs_vtxreweight0_extract.py` and `/tmp/extract_mbd_and` (not committed).